



Review article

Early menopause and premature ovarian insufficiency are associated with increased risk of dementia: A systematic review and meta-analysis of observational studies

Eleni K. Karamitrou^a, Panagiotis Anagnostis^{a,*}, Konstantina Vaitis^a, Loukas Athanasiadis^b,
Dimitrios G. Goulis^a

^a Unit of Reproductive Endocrinology, 1st Department of Obstetrics and Gynecology, Medical School, Aristotle University of Thessaloniki, Thessaloniki, Greece

^b First Department of Psychiatry, Medical School, Aristotle University of Thessaloniki, Greece

ARTICLE INFO

Keywords:

Early menopause
Premature ovarian insufficiency
Dementia
Alzheimer's disease
Cognitive function

ABSTRACT

Background/aims: Among other risk factors, the decline in estrogen concentrations during menopause may compromise cognitive function. Whether early menopause (EM) is associated with an increased risk of dementia remains unclear. The purpose of this study was to systematically review and meta-analyze current evidence regarding the association between EM or premature ovarian insufficiency (POI) and the risk of dementia of any type.

Materials and methods: A comprehensive literature search was conducted through the PubMed, Scopus and CENTRAL databases up to August 2022. Study quality was assessed using the Newcastle-Ottawa scale. Associations were calculated as odds ratio (OR) with 95 % confidence interval (CI). The I^2 index was employed for heterogeneity.

Results: Eleven studies (nine assessed as of good and two as of fair quality) were included in the meta-analysis ($n = 4,716,862$). Women with EM demonstrated a greater risk of dementia of any type than women of normal age at menopause (OR 1.37, 95 % CI 1.22–1.54; I^2 93%). However, after excluding a large retrospective cohort study, the results were altered (OR 1.07, 95 % CI 0.78–1.48; I^2 94%). Increased risk of dementia was also found in women with POI (OR 1.18, 95 % CI 1.15–1.21; I^2 0%). Subgroup analysis showed that this risk was mostly evident in cohort studies, and those which included women with natural menopause.

Conclusions: Women with EM or POI may be at increased risk of dementia compared with women of normal age at menopause, but further research investigating that hypothesis is warranted.

1. Introduction

Menopause is the permanent end of menstruation induced by follicle depletion [1]. The age range of natural menopause is 48–52 years [1]. However, in 10 % of cases, menopause may occur before the age of 45 years, defined as early menopause (EM), whereas in 1 % of cases, the age at menopause is younger than 40 years, defined as premature ovarian insufficiency (POI) [1]. Both natural and iatrogenic causes of EM, such as chemotherapy, radiotherapy or surgery (e.g., bilateral oophorectomy), have been identified [2]. The end of follicle lifespan, which marks the entry into the postmenopausal period, is determined by a complicated and not yet fully mapped interaction among genetic, hormonal and environmental factors [3].

The term “dementia” refers to a major neurocognitive disorder characterized by cognitive impairment and gradual occupational and social functioning decline [4]. Cognitive deficits may be seen in many mental disorders [4]. However, the Neurocognitive Disorder Category includes only those whose core features are cognitive [4]. According to the Diagnostic and Statistical Manual of Mental Disorders (5th ed.; DSM-V) criteria, the loss of at least one cognitive function (regarding memory or other domains, such as language), which is accompanied by a decline from a previously established level of cognitive function and impairment of daily-life-processes, such as social and professional obligations or self-care procedures, defines dementia [4]. Fifty-seven million people have dementia worldwide, and its prevalence is expected to triple by 2050 [5]. Alzheimer's Disease (AD) is the most common cause of dementia

* Corresponding author.

E-mail address: pan.anagnostis@gmail.com (P. Anagnostis).

<https://doi.org/10.1016/j.maturitas.2023.107792>

Received 22 December 2022; Received in revised form 6 June 2023; Accepted 17 June 2023

Available online 23 June 2023

0378-5122/© 2023 Elsevier B.V. All rights reserved.

[6]. Other types of dementia involve vascular dementia, Lewy body dementia and conditions, such as frontotemporal dementia, progressive aphasia, Parkinson's disease dementia, and secondary dementia [6]. The loss of estrogen's neuroprotective action has been associated with neuroinflammation, synaptic decline and cognitive impairment, potentially leading to dementia development [7]. In addition, factors other than estrogen deficiency, like increased iron concentrations [8], which are also associated with age at menopause, may be implicated in the mechanisms that lead to cognitive impairment.

However, it remains unclear whether the effect of estrogen on the brain is age-dependent and whether the risk of dementia is associated with age during the transition to menopause [3]. Few observational studies have investigated the association between EM or POI and dementia yielding contradictory results [9–13].

This systematic review and meta-analysis aimed to systematically review and meta-analyze the existing evidence regarding the association between non-genetic EM (natural or surgical) and the risk of all-cause dementia.

2. Materials and methods

2.1. Guidelines followed

Our study was conducted according to the published work by Muka et al. [14] and was reported following the MOOSE (Meta-analyses Of Observational Studies in Epidemiology) guidelines [15] (Supplementary Table 1). The flow chart is presented in Fig. 1. The present study has also

been registered in the Prospective Register of Systematic Reviews (PROSPERO) System (PROSPERO ID: CRD42022285762).

2.2. Search strategy

To identify relevant studies, a systematic literature search was carried out in electronic databases MedLine, Scopus, and CENTRAL (Cochrane), from the beginning of registration up to August 14, 2022. Indicatively, search string used in MedLine was as follows: ("menopause, premature"[MeSH] OR "primary ovarian insufficiency"[MeSH] OR "ovarian insufficiency"[tiab] OR "ovarian failure"[tiab] OR ((menopause[MeSH] OR menopause[tiab] OR menopausal[tiab] OR climacteric[tiab] OR postmenopausal[tiab] OR post-menopausal[tiab])) AND (early[tiab] OR premature[tiab] OR age[tiab] OR years[tiab] OR time[tiab])) AND (dementia[MeSH] OR dementia[tiab] OR "Alzheimer Disease"[MeSH] OR Alzheimer[tiab] OR Alzheimer's[tiab] OR Alzheimers [tiab] OR "AD"[tiab] OR "cognitive dysfunction"[MeSH] OR "cognitive dysfunction"[tiab] OR "impaired cognition"[tiab] OR "cognitive deficit"[tiab] OR "cognitive disorders"[tiab] OR "cognitive disorder"[tiab] OR "healthy brain"[tiab] OR "brain health"[tiab] OR "Vascular Dementia"[MeSH] OR "vascular dementia"[tiab] OR "VaD"[tiab] OR "Frontotemporal Lobar Degeneration"[MeSH] OR "frontotemporal lobar degeneration"[tiab] OR "neurocognitive disorders"[MeSH] OR "neurocognitive disorders"[tiab] OR "neurocognitive disorder"[tiab] OR "neurocognitive impairment"[tiab] OR "neurocognitive deficit"[tiab] OR "neurocognitive dysfunction"[tiab] OR "impaired neurocognition"[tiab] OR "impaired cognition"[tiab] OR

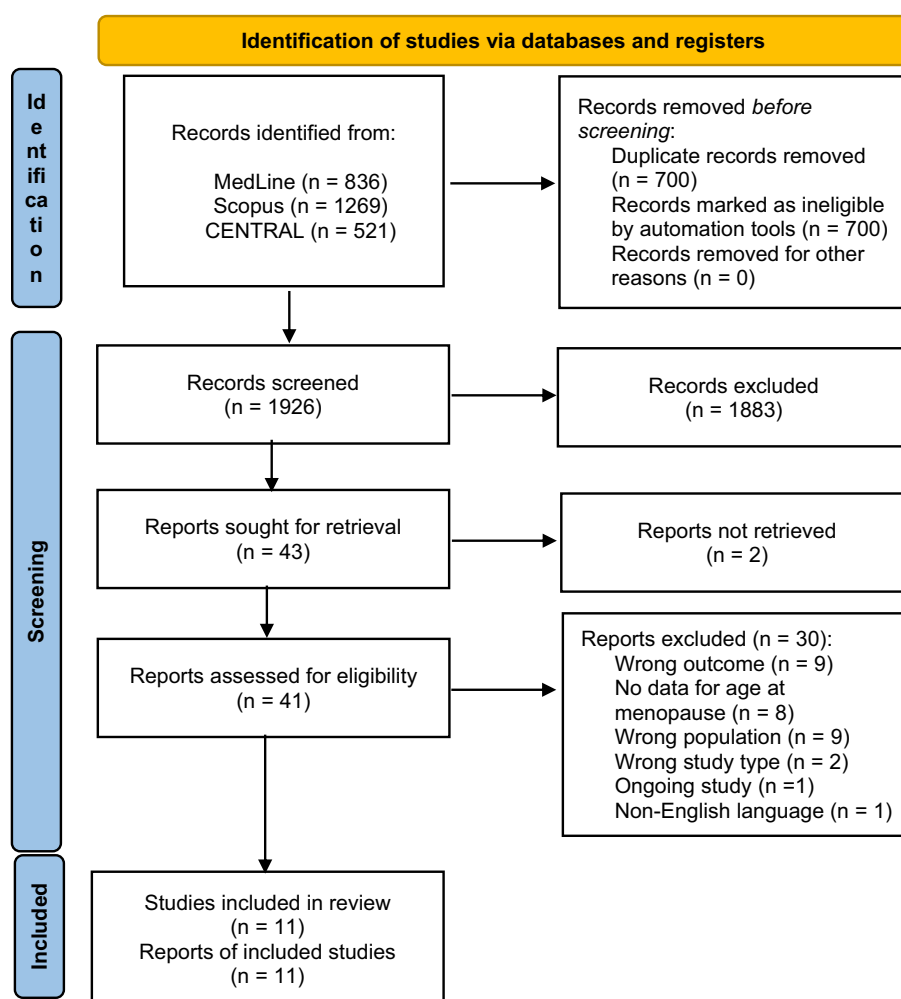


Fig. 1. Flow chart diagram.

"Mild Cognitive Impairment"[MeSH] OR "mild cognitive impairment"[tiab] OR "MCI"[tiab] OR "Mental Status and Dementia Tests"[MeSH] OR "Mini-Mental State Examination"[tiab] OR "MMSE"[tiab]) NOT (Animal[MeSH] NOT Human[MeSH]) NOT (letter[pt] OR comment[pt] OR editorial[pt] OR Review[pt] OR "practice guideline"[ptyp] OR "case reports"[ptyp]).

The online tool *Rayyan* (Intelligent Systematic Review) was used to screen studies and remove duplicate reports. In brief, all studies resulting from the literature search were examined at the title and abstract levels. In the second phase, studies were screened at a full-text level for inclusion criteria. Data were subsequently extracted. Two independent researchers (EKK and KV) carried out the above procedure. Disagreements were resolved by a third independent researcher (PA). Both cohort and case-control studies were considered eligible. Grey literature, such as conference abstracts, was also searched for eligible reports. References of included studies were screened for eligibility. When necessary, an e-mail was sent to the corresponding author to retrieve data.

2.3. Study selection

The PECO (Population, Exposure, Comparator, Outcome) components that were used to conduct our systematic review were as follows: (i) Population: postmenopausal women; (ii) Exposure: age at menopause <45 years (either EM or POI: age at menopause <40 years); (iii) Comparator: women of normal age at menopause (>45 years); (iv) Outcome: dementia from any cause.

Study exclusion criteria were: (i) lack of control group; (ii) inclusion of premenopausal or perimenopausal women; (iii) inclusion of patients with genetic syndromes associated with EM or POI (e.g., Turner's syndrome); (iv) inclusion of patients with a history of the polycystic ovarian syndrome. Animal studies, reviews, clinical trials, case reports, abstracts, case series, letters-to-the editor and articles not written in English were also excluded. Cohort, case-control and cross-sectional studies were included.

2.4. Data extraction

The data of the included studies were recorded into pre-designed extraction forms. The following information was recorded for each study: (i) name of the first author; (ii) year of publication; (iii) country of study; (iv) type of study; (v) duration of the study; (vi) sample size; (vii) mean age of participants; (viii) type of menopause (surgical or natural); (ix.) mean age of menopause (x) administration of menopausal hormone therapy (MHT); (xi) number of participants with normal age at menopause (>45 years); (xii) the number of participants with EM or POI (<40 years); (xiii) the number of participants with dementia in each of the above categories. Variables, such as level of education, presence of chronic diseases, smoking, alcohol use, exercise, diet, and the number of pregnancies, were also included.

The incidence of dementia was compared between women with EM or POI and women with a history of normal age at menopause (>45 years). In studies in which groups of postmenopausal women were formed using different age thresholds (and not 45 years of age), the most appropriate group close to this threshold was used on a case-by-case basis.

2.5. Risk of bias and study quality assessment

Quality assessment of the studies was performed according to the Newcastle-Ottawa scale (NOS), based on three main criteria: (i) selection of study groups; (ii) comparability of groups; (iii) assessment of exposure or outcome (for case-control or cohort studies, respectively) [16]. Each category was evaluated based on a four, two and three-star scale (the maximum), respectively (total maximum score: nine stars). A study was awarded as of "good quality" when it gets 3–4 stars in the

selection domain and 1–2 stars in the comparability domain, and 2–3 stars in the outcome/exposure domain. "Fair quality" is considered when two stars are obtained in the selection domain and 1–2 stars in the comparability domain, and 2–3 stars in the outcome/exposure domain. Finally, a study is characterized as of "poor quality" in the case of 0–1 stars in the selection domain or 0 stars in the comparability domain or 0–1 stars in the outcome/exposure domain [16] Supplementary Table 2.

2.6. Statistical analysis

The association between the risk factor (EM or POI) and outcome (dementia) is presented as odds ratio (OR) with a 95 % confidence interval (CI), with p values <0.05 being considered statistically significant. An inverse variance random effects meta-analysis was performed. Moreover, the heterogeneity of studies was examined using the Cochrane chi-square test and was quantified by I^2 statistics. I^2 values <40 % were considered a "low degree" of heterogeneity, whereas I^2 values 40–60 % and > 60 % were evaluated as "moderate" and "high degree" of heterogeneity. Sensitivity and subgroup analyses were also performed to reduce heterogeneity. Furthermore, Funnel plots and Egger's test were used to determine the likelihood of publication bias. All analyses were conducted using the *Review Manager (RevMan 5.4.1)* software) computer program (Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration, September 2020).

3. Results

3.1. Descriptive data

The initial database search identified 2626 studies after removing duplicates, 42 of which were ultimately assessed at the full-text level (two studies could not be retrieved). A total of 11 studies [11,17–26] were eligible for quantitative synthesis (meta-analysis) (Fig. 1). The exclusion reasons for the remaining 31 studies are listed in Supplementary Table 3.

Six of the selected studies were cohorts [one retrospective [21] and five prospective [17,18,23,24,26]] and five case-control [11,22,27]. The publication period of the included studies ranged from 1994 to 2022. Five studies were conducted in the USA [11,17,19,23,25], one in South Korea [21], one in Italy [22], one in China [24], one in Sweden [20], one in French [26] and one in the UK [18]. The total number of participants meeting our inclusion criteria was 4,716,862. EM was diagnosed in 328,497 women, 326,508 of whom were diagnosed with natural menopause, 636 with surgical menopause, and the remaining 1078 with either natural or surgical menopause (no data separating the two conditions were available). Information about the average age at menopause was provided in one study [21], and data about MHT use were provided in seven studies [17–22,24]. The descriptive characteristics of the included studies are presented in Table 1. Heterogeneity was observed for the categorization according to age at menopause in each study and for the definition of dementia.

3.2. Risk of bias

The quality assessment of included studies is illustrated in Supplementary Table 2. All cohort studies were assessed as of "low risk of bias" [17,18,21,23,24,26]. Three of the case-control studies were also assessed as of "low risk of bias" [20,22,25], while two were assessed as of "medium risk of bias" [11,19].

3.3. Main analysis

Women with EM showed a greater risk of dementia than those with normal age at menopause (OR 1.37, 95 % CI 95 % 1.22–1.54; I^2 93%, Tau^2 0.02, Chi^2 138.48, df 10) (Fig. 2).

Table 1
Demographic characteristics of studies included in the meta-analysis.

ID	First author, year of publication	Country	Study design	Participants (n)	Mean age (years)	Type of menopause	Menopause age (years)	MHT use (%)
1	Paganini-Hill, 1994	USA	Case-control	652	N/A	Natural and surgical	N/A	N/A
2	Rasgon, 2005	Sweden	Case-control	4461	N/A	Natural and surgical	N/A	N/A
3	Yoo, 2020	North Korea	Cohort (retrospective)	4,696,633	61.2 ± 8.6	Natural	50.2 ± 4.0	18.4 %
4	Rocca, 2007	USA	Case-control	473	N/A	Surgical	N/A	N/A
5	Zucchella, 2012	Italy	Case-control	551	Cases: 77.6 ± 6.3 Controls: 76.7 ± 7.5	Natural and surgical	N/A	6.9 %
6	Ryan, 2014	French	Cohort (prospective)	2997	N/A	Natural and surgical	N/A	N/A
7	Gilsanz, 2019	USA	Cohort (prospective)	1645	N/A	Natural	N/A	N/A
8	Paganini-Hill, 2020	USA	Cohort (prospective)	420	96.5 ± 3.2	Natural and surgical	N/A	70 %
9	Song, 2020	China	Cohort (prospective)	8222	73.6 ± 6.6	Natural	N/A	9 %
10	Rocca, 2021	USA	Case-control	386	N/A	Surgical	N/A	42.9 %
11	Gong, 2022	UK	Cohort (prospective)	422	N/A	Natural	N/A	N/A

Data are presented as mean ± SD or n (%). Abbreviations: MHT: menopausal hormone therapy; N/A not available.

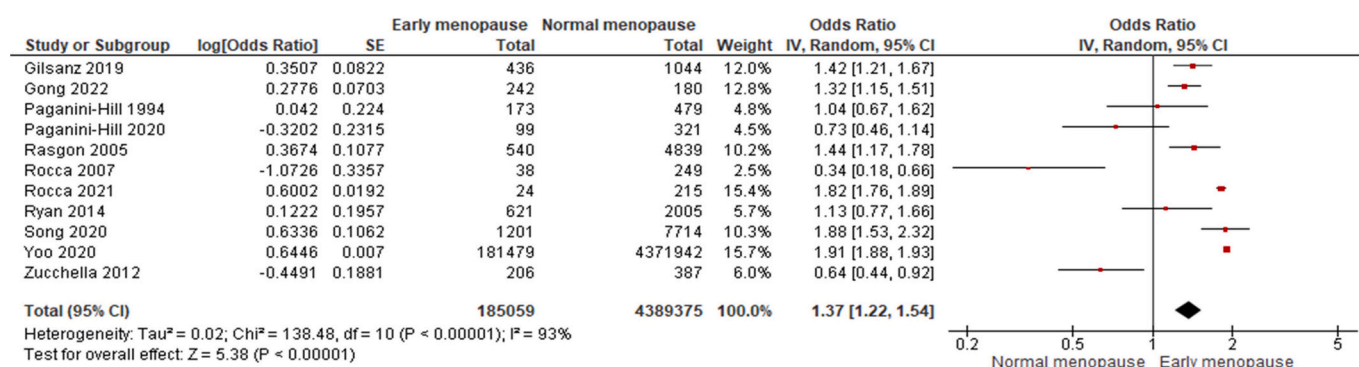


Fig. 2. Forest plot of the comparison between early menopause and menopause at a normal age.

3.4. Sensitivity analysis

After excluding the studies by Gilsanz et al. 2019 [23], Gong et al. 2022 [18], Rocca et al. 2007 [11], Ryan et al. 2014 [26] and Song et al. 2020 [24], which did not classify participants in accordance to the criteria used in our study (different age cut-off values), the results were not altered (OR 1.49, 95 % CI 1.31–1.68; I^2 93%, τ^2 0.01, χ^2 68.58, df 5). However, a separate analysis after excluding the study by Yoo et al. 2020 [21], which included a very large sample size, did alter our results (OR 1.07, 95 % CI 0.78–1.48; I^2 94%, τ^2 0.09, χ^2 106.44, df 9). Moreover, a separate analysis after excluding the study by Paganini-Hill et al. 2020 [17], which examined the incidence of dementia in participants >90 years of age (a population with a confirmed greater risk of dementia), did not alter our results (OR 1.55, 95 % CI 1.38–1.75; I^2 93%, τ^2 0.02, χ^2 121.63, df 9). Finally, a sensitivity analysis based on the quality assessment of the studies was performed. In an analysis excluding the two studies (Paganini-Hill et al. 1994 [19] and Rocca et al. 2007 [11]) assessed as of “fair quality”, an association between EM and dementia was still significant (OR 1.47, 95 % CI 1.32–1.63; I^2 92%, τ^2 0.02, χ^2 105.85, df 8).

3.5. Subgroup analysis

Subgroup analysis according to study design showed an increased risk of dementia in the EM population only in cohorts (OR 1.41, 95 % CI 1.13–1.76; I^2 92%, τ^2 0.06, χ^2 63.46, df 3) but not in case-control studies (OR 0.97, 95 % CI 0.62–1.52; I^2 94%, τ^2 0.11, χ^2 18.06,

df 4). This result remained even after performing the same analysis, by excluding the large retrospective cohort study by Yoo et al. 2020 [21], which could have biased the results (OR 1.32, 95 % CI 1.07–1.63; I^2 78%, τ^2 0.04, χ^2 17.80, df 4) (Fig. 3).

Subgroup analysis according to the type of menopause (surgical or natural) showed an increased risk of dementia only in women with natural EM menopause (OR 1.61, 95 % CI 1.30–2.00; I^2 92%, τ^2 0.04, χ^2 39.37, df 3). When the analysis was confined to women with POI, an increased risk of dementia was also observed compared to women of normal age at menopause (OR 1.18, 95 % CI 1.15–1.21; I^2 0%, χ^2 2.59, df 3) (Fig. 4).

Subgroup analyses according to MHT use, education level, presence of chronic diseases, smoking, alcohol use, exercise, diet, and the number of pregnancies were not performed due to lack of extractable data in the included studies.

4. Discussion

To our knowledge, the present study is the largest systematic review and meta-analysis (including almost five million participants), which examined the association between age at menopause and dementia. Women with EM or POI may be at increased risk of dementia compared to women with age at menopause >45 years, but further research investigating that hypothesis is warranted.

A meta-analysis by Fu et al., published in 2021 [28], including 10 studies, which reported results for the same comparison with our study, showed that menopause after 45 years of age was consistently associated

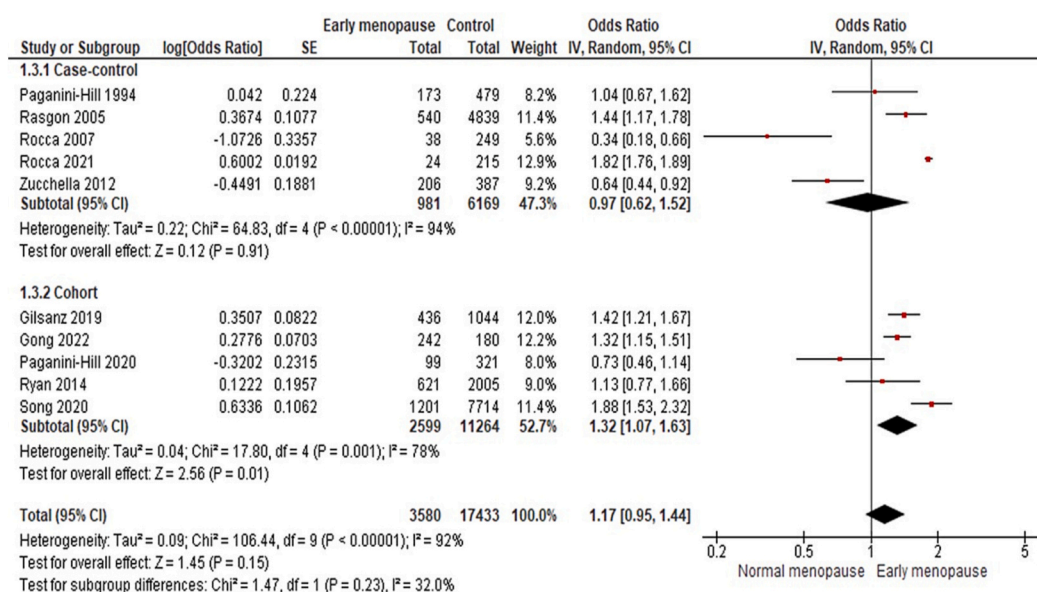


Fig. 3. Forest plot of the subgroup analysis based on the type of study (case-control or cohort, excluding the large retrospective cohort study).

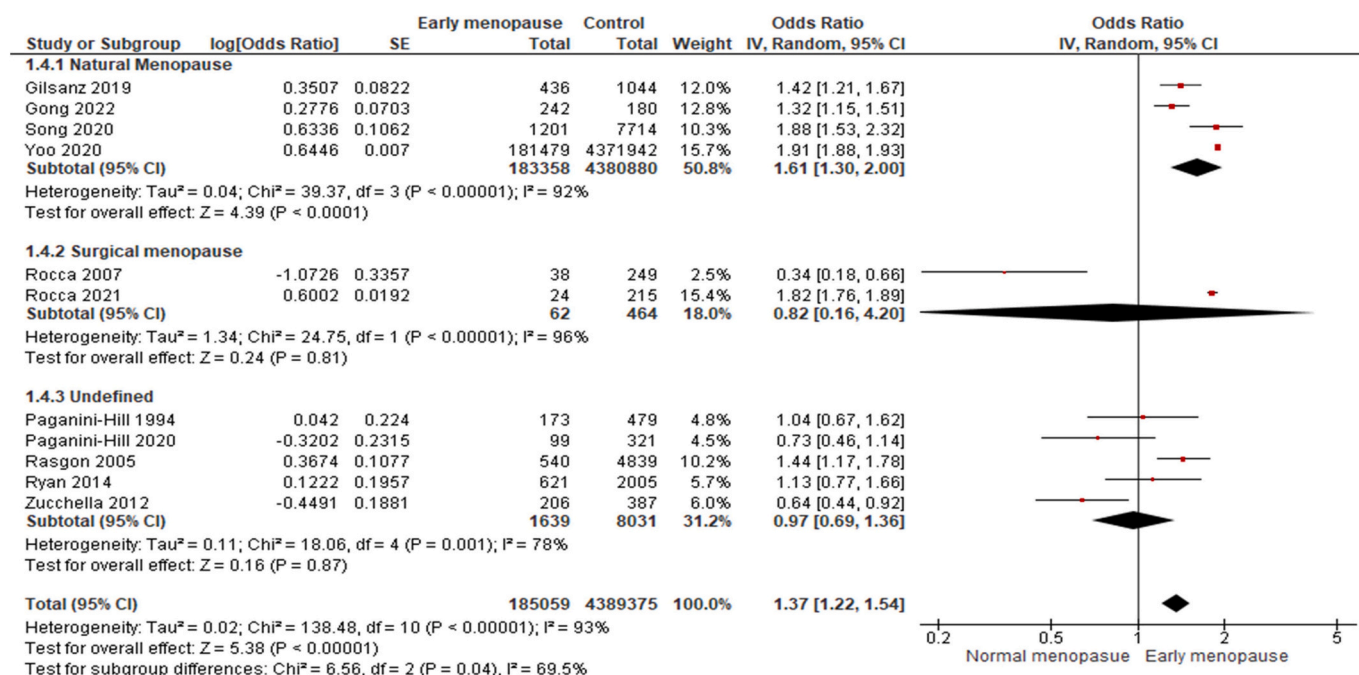


Fig. 4. Forest plot of the subgroup analysis based on the type of menopause (surgical or natural).

with a lower risk of all-cause dementia (pooled relative risk: 0.87, 95 % CI 0.78–0.95, I^2 56%) compared to women with menopause at age < 45 years. However, no discrimination regarding the type of menopause (natural or surgical) and type of premature menopause (EM or POI) was reported. Such analysis was performed in our study. Moreover, there were differences in the populations included (e.g., exclusion of patients diagnosed with Down syndrome), databases searched and inclusion-exclusion criteria.

Estrogens affect areas of the central nervous system (CNS) that control learning, registration and retrieval of information, judgment, evaluation and speech [29]. These regions are mainly located in the prefrontal cortex, hippocampus and striatum [29]. An estrogen-mediated increase in cellular proteins that promote neuronal growth, survival, and function, as well as neurotransmission and synaptogenesis,

has been reported [29]. In addition, estrogens have been shown to limit inflammatory responses in CNS, thus effectively protecting it against dystrophy and dementia [30]. In addition, factors other than estrogen deficiency, like increased iron concentrations [8], which are also associated with menopause may be implicated in the mechanisms that lead to cognitive impairment. Nevertheless, mounting evidence supports the effect of menopause on cognitive functions, mainly on verbal memory and verbal fluency [30].

Whether MHT protects neurological functions after menopause at any age remains controversial [31]. In older women (>60 years), health risks appear to outweigh the benefits, and, therefore, MHT is not recommended in this population [31]. In the Women's Health Initiative study, postmenopausal women using MHT presented an increased risk of dementia compared with the placebo group (hazard ratio 2.05, 95 % CI

1.21–3.48) [31]. Interestingly, in a meta-analysis by Zhou et al. 2021, which examined the effect of hormone replacement therapy (HRT) on cognitive function in healthy postmenopausal women, it was found that HRT had a negative effect on global cognition (standardized mean difference (SMD) -0.04 , 95 % CI -0.08 to -0.01) [32]. A recent meta-analysis by Song et al., published in 2020, showed that estrogen replacement therapy decreased the risk of developing AD (OR 0.672; 95 % CI 0.581–0.779; $P < 0.001$), with factors, such as hormone therapy ascertainment ($p = 0.494$) and duration of treatment ($p = 0.217$), not affecting the association between MHT and AD risk [33]. However, the issue of whether younger women with POI could benefit from estrogen therapy, which could protect them from cognitive decline and increased dementia risk in later life, remains unclear [34,35]. Although a direct comparison of the estimate of risk for dementia between women with EM receiving MHT compared to women with normal menopausal age would be of particular interest, data provided by the included studies were insufficient to facilitate its conduct.

The present study has some limitations. First, increased heterogeneity was observed among studies, attributed to a different definition of EM and dementia, variation in the population studied, interventions and outcomes measured, which have compromised the validity of our results. Second, no sufficient data were extractable to perform subgroup analyses according to the type of dementia or the role of important dementia-related factors, such as patients' age, education level, MHT use, smoking, diet, and alcohol abuse. Although, some studies provided adjusted measures of effect for many of the above variables, the pre-defined patient subgroups according to age did not match that of our research question. To appropriately utilize the given data and conduct meaningful syntheses, we used data given as proportion of participants. Third, the findings of this meta-analysis were largely affected by a sensitivity analysis, which excluded the large retrospective study. Fourth, the design of included studies also affected the validity of our results, since the association between EM and dementia was significant only in cohort studies.

In conclusion, this meta-analysis shows that POI and EM may be associated with an increased risk of dementia. While our study is indicative, it cannot provide strong evidence of such association, due to high heterogeneity and relatively small number of the studies included. To investigate the association between EM and dementia future observational studies should be conducted. The pathophysiological mechanisms for such an interplay need also further investigation through experimental studies. In any case, there is also an urgent need for RCTs focusing on the use of MHT or other lifestyle interventions to avert the development of dementia in this subpopulation of postmenopausal women.

Contributors

Eleni K. Karamitrou analyzed the extractable data, was responsible for the statistical analysis and wrote the first draft of the paper.

Panagiotis Anagnostis designed the research, reviewed the manuscript, resolved discrepancies, and provided critical scientific input.

Konstantina Vaitis searched the literature and extracted the data.

Loukas Athanasiadis reviewed the manuscript and provided critical scientific input.

Dimitrios G. Goulis reviewed the manuscript, provided critical scientific input and was primarily responsible for the paper's final content.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Provenance and peer review

This article was not commissioned and was externally peer reviewed.

Declaration of competing interest

The authors declare that they have no competing interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.maturitas.2023.107792>.

References

- [1] S.E. Tom, G.D. Mishra, *Current Topics in Menopause: A Life Course Approach to Reproductive Aging*, Bentham Science Publishers Ltd., 2013.
- [2] E.L. Scott, Q.-G. Zhang, R.K. Vadlamudi, D.W. Brann, Premature menopause and risk of neurological disease: basic mechanisms and clinical implications, *Mol. Cell. Endocrinol.* 389 (2014) 2–6, <https://doi.org/10.1016/j.mce.2014.01.013>.
- [3] S.R. Davis, I. Lambrinoudaki, M. Lumsden, G.D. Mishra, L. Pal, M. Rees, N. Santoro, T. Simoncini, Menopause, *Nat. Rev. Dis. Prim.* 1 (2015) 15004, <https://doi.org/10.1038/nrdp.2015.4>.
- [4] *Diagnostic and Statistical Manual of Mental Disorders*, American Psychiatric Association, Washington, DC, 5th ed, 2013.
- [5] E. Nichols, J.D. Steinmetz, S.E. Vollset, K. Fukutaki, J. Chalek, F. Abd-Allah, A. Abdoli, A. Abualhasan, E. Abu-Gharbieh, T.T. Akram, H. Al Hamad, F. Alahdab, F.M. Alanezi, V. Alipour, S. Almustanyir, H. Amu, I. Ansari, J. Arabloo, T. Ashraf, T. Astell-Burt, G. Ayano, J.L. Ayuso-Mateos, A.A. Baig, A. Barnett, A. Barrow, B. T. Baune, Y. Béjot, W.M.M. Bezabhe, Y.M. Bezabih, A.S. Bhagavathula, S. Bhaskar, K. Bhattacharyya, A. Bijani, A. Biswas, S.R. Bolla, A. Boloor, C. Brayne, H. Brenner, K. Burkart, R.A. Burns, L.A. Cámara, C. Cao, F. Carvalho, L.F.S. Castro-de-Araujo, F. Catalá-López, E. Cerin, P.P. Chavan, N. Cherbuin, D.-T. Chu, V.M. Costa, R.A. S. Couto, O. Dadrás, X. Dai, L. Dandona, R. Dandona, V. De la Cruz-Góngora, D. Dhamnetiya, D. Dias da Silva, D. Diaz, A. Douiri, D. Edvardsson, M. Ekholuenetale, I. El Sayed, S.I. El-Jaafari, K. Eskandari, S. Eskandarieh, S. Esmailnejad, J. Fares, A. Faro, U. Farooque, V.L. Feigin, X. Feng, S.-M. Fereshtehnejad, E. Fernandes, P. Ferrara, I. Filip, H. Fillit, F. Fischer, S. Gaidhane, L. Galluzzo, A. Ghashghaee, N. Ghith, A. Gialluisi, S.A. Gilani, I.-R. Glavan, E.V. Gnedovskaya, M. Golechha, R. Gupta, V.B. Gupta, V.K. Gupta, M. R. Haider, B.J. Hall, S. Hamidi, A. Hanif, G.J. Hankey, S. Haque, R.K. Hartono, A. I. Hasaballah, M.T. Hasan, A. Hassan, S.I. Hay, K. Hayat, M.I. Hegazy, G. Heidari, R. Heidari-Soureshjani, C. Herteliu, M. Househ, R. Hussain, B.-F. Hwang, L. Iacoviello, I. Iavicoli, O.S. Ilesanmi, I.M. Ilic, M.D. Ilic, S.S.N. Irvani, H. Iso, M. Iwagami, R. Jabbarinejad, L. Jacob, V. Jain, S.K. Jayapal, R. Jayawardena, R. P. Jha, J.B. Jonas, N. Joseph, R. Kalani, A. Kandel, H. Kandel, A. Karch, A.S. Kasa, G.M. Kassie, P. Keshavarz, M.A.B. Khan, M.N. Khatib, T.A.M. Khoja, J. Khubchandani, M.S. Kim, Y.J. Kim, A. Kisa, S. Kisa, M. Kivimäki, W.J. Koroshetz, A. Koyanagi, G.A. Kumar, M. Kumar, M.H. Lak, M. Leonardi, B. Li, S.S. Lim, X. Liu, Y. Liu, G. Logroscino, S. Lorkowski, G. Lucchetti, R. Lutzky Saute, F.G. Magnani, A. A. Malik, J. Massano, M.M. Mehndiratta, R.G. Menezes, A. Meretoja, B. Mohajer, N. Mohamed Ibrahim, Y. Mohammad, A. Mohammed, A.H. Mokdad, S. Mondello, M. A.A. Moni, M. Moniruzzaman, T.B. Mossie, G. Nagel, M. Naveed, V.C. Nayak, S. Neupane Kandel, T.H. Nguyen, B. Oancea, N. Ostavnov, S.S. Ostavnov, M. O. Owolabi, S. Panda-Jonas, F. Pashazadeh Kan, M. Pasovic, U.K. Patel, M. Pathak, M.F.P. Peres, A. Perianayagam, C.B. Peterson, M.R. Phillips, M. Pinheiro, M. A. Piradov, C.D. Pond, M.H. Potashman, F.H. Pottoo, S.I. Prada, A. Radfar, A. Raggi, F. Rahim, M. Rahman, P. Ram, P. Ranasinghe, D.L. Rawaf, S. Rawaf, N. Rezaei, A. Rezapour, S.R. Robinson, M. Romoli, G. Roshandel, R. Sahathevan, A. Sahebkar, M.A. Sahraian, B. Sathian, D. Sattin, M. Sawhney, M. Saylan, S. Schiavolin, A. Seylani, F. Sha, M.A. Shaikh, K.S. Shaji, M. Shannawaz, J. K. Shetty, M. Shigematsu, J. Il Shin, R. Shiri, D.A.S. Silva, J.P. Silva, R. Silva, J. A. Singh, V.Y. Skryabin, A.A. Skryabina, A.E. Smith, S. Soshnikov, E.E. Spurlock, D. J. Stein, J. Sun, R. Tabarés-Seisdedos, B. Thakur, B. Timalina, M.R. Tovani-Palone, B.X. Tran, G.W. Tsegaye, S. Valadan Tahbaz, P.R. Valdez, N. Venketasubramanian, V. Vlassov, G.T. Vu, L.G. Vu, Y.-P. Wang, A. Wimo, A.S. Winkler, L. Yadav, S.H. Yahyazadeh Jabbari, K. Yamagishi, L. Yang, Y. Yano, N. Yonemoto, C. Yu, I. Yunusa, S. Zadey, M.S. Zastroszhin, A. Zastroszhina, Z.-J. Zhang, C.J.L. Murray, T. Vos, Estimation of the global prevalence of dementia in 2019 and forecasted prevalence in 2050: an analysis for the Global Burden of Disease Study 2019, *Lancet Public Health* 7 (2022) e105–e125, [https://doi.org/10.1016/S2468-2667\(21\)00249-8](https://doi.org/10.1016/S2468-2667(21)00249-8).
- [6] K. Lam, W.S.Y. Chan, J.K.H. Luk, A.Y.M. Leung, Assessment and diagnosis of dementia: a review for primary healthcare professionals, *Hong Kong Med. J.* 25 (2019) 473–482, <https://doi.org/10.12809/hkmj198073>.
- [7] S. Zárte, T. Stevnsner, R. Gredilla, Role of estrogen and other sex hormones in brain aging. Neuroprotection and DNA repair, *Front. Aging Neurosci.* 9 (2017) 430, <https://doi.org/10.3389/fnagi.2017.00430>.
- [8] J. Jian, E. Pelle, X. Huang, Iron and menopause: does increased iron affect the health of postmenopausal women? *Antioxid. Redox Signal.* 11 (2009) 2939–2943, <https://doi.org/10.1089/ars.2009.2576>.
- [9] X. Hong, X. Zhang, H. Li, A case-control study of endogenous estrogen and risk of Alzheimer's disease, *Zhonghua Liu Xing Bing Xue Za Zhi* 22 (2001) 379–382.
- [10] R.N. McLay, P.M. Maki, C.G. Lyketsos, Nulliparity and late menopause are associated with decreased cognitive decline, *J. Neuropsychiatr. Clin. Neurosci.* 15 (2003) 161–167, <https://doi.org/10.1176/jnp.15.2.161>.

- [11] W.A. Rocca, J.H. Bower, D.M. Maraganore, J.E. Ahlskog, B.R. Grossardt, M. de Andrade, L.J. Melton 3rd, Increased risk of cognitive impairment or dementia in women who underwent oophorectomy before menopause, *Neurology*. 69 (2007) 1074–1083, <https://doi.org/10.1212/01.wnl.0000276984.19542.e6>.
- [12] T.K.T. Phung, B.L. Waltoft, T.M. Laursen, A. Settnes, L.V. Kessing, P.B. Mortensen, G. Waldemar, Hysterectomy, oophorectomy and risk of dementia: a nationwide historical cohort study, *Dement. Geriatr. Cogn. Disord.* 30 (2010) 43–50, <https://doi.org/10.1159/000314681>.
- [13] R. Bove, E. Secor, L.B. Chibnik, L.L. Barnes, J.A. Schneider, D.A. Bennett, P.L. De Jager, Age at surgical menopause influences cognitive decline and Alzheimer pathology in older women, *Neurology*. 82 (2014) 222–229, <https://doi.org/10.1212/WNL.0000000000000033>.
- [14] T. Muka, M. Glisic, J. Milic, S. Verhoog, J. Bohlius, W. Bramer, R. Chowdhury, O. H. Franco, A 24-step guide on how to design, conduct, and successfully publish a systematic review and meta-analysis in medical research, *Eur. J. Epidemiol.* 35 (2020) 49–60, <https://doi.org/10.1007/s10654-019-00576-5>.
- [15] D.F. Stroup, J.A. Berlin, S.C. Morton, I. Olkin, G.D. Williamson, D. Rennie, D. Moher, B.J. Becker, T.A. Sipe, S.B. Thacker, Meta-analysis of observational studies in epidemiology: a proposal for reporting. Meta-analysis Of Observational Studies in Epidemiology (MOOSE) group, *JAMA* 283 (2000) 2008–2012, <https://doi.org/10.1001/jama.283.15.2008>.
- [16] G.A. Wells, G. Wells, B. Shea, B. Shea, D. O'Connell, J. Peterson, M. Losos Welch, P. Tugwell, S.W. Ga, G.A. Zello, J.A. Petersen, The Newcastle-Ottawa Scale (NOS) for Assessing the Quality of Nonrandomised Studies in Meta-Analyses, 2014.
- [17] A. Paganini-Hill, M.M. Corrada, C.H. Kawan, Prior endogenous and exogenous estrogen and incident dementia in the 10th decade of life: the 90+ study, *Climacteric*. 23 (2020) 311–315, <https://doi.org/10.1080/13697137.2020.1727876>.
- [18] J. Gong, K. Harris, S.A.E. Peters, M. Woodward, Reproductive factors and the risk of incident dementia: a cohort study of UK Biobank participants, *PLoS Med.* 19 (2022) 1–23, <https://doi.org/10.1371/journal.pmed.1003955>.
- [19] A. Paganini-hill, V.W. Henderson, Estrogen deficiency and risk of alzheimer's disease in women, *Am. J. Epidemiol.* 140 (1994) 256–261, <https://doi.org/10.1093/oxfordjournals.aje.a117244>.
- [20] N.L. Rasgon, C. Magnusson, A.L.V. Johansson, N.L. Pedersen, S. Elman, M. Gatz, Endogenous and exogenous hormone exposure and risk of cognitive impairment in Swedish twins: a preliminary study, *Psychoneuroendocrinology*. 30 (2005) 558–567, <https://doi.org/10.1016/j.psyneuen.2005.01.004>.
- [21] J.E. Yoo, D.W. Shin, K. Han, D. Kim, H.-S. Won, J. Lee, S.Y. Kim, G.E. Nam, H. S. Park, Female reproductive factors and the risk of dementia: a nationwide cohort study, *Eur. J. Neurol.* 27 (2020) 1448–1458, <https://doi.org/10.1111/ene.14315>.
- [22] C. Zucchella, E. Sinforiani, A. Citterio, V. Giarracca, G. Bono, M. Mauri, Reproductive life events and Alzheimer's disease in Italian women: a retrospective study, *Neuropsychiatr. Dis. Treat.* 8 (2012) 555–560, <https://doi.org/10.2147/NDT.S36622>.
- [23] P. Gilsanz, C. Lee, M.M. Corrada, C.H. Kawan, C.P. Quesenberry, R.A. Whitmer, Reproductive period and risk of dementia in a diverse cohort of health care members, *Neurology* 92 (2019) E2005–E2014, <https://doi.org/10.1212/WNL.0000000000007326>.
- [24] X. Song, J. Wu, Y. Zhou, L. Feng, J.M. Yuan, A. Pan, W.P. Koh, Reproductive and hormonal factors and risk of cognitive impairment among Singapore Chinese women, *Am. J. Obstet. Gynecol.* 223 (410) (2020) e1–410.e23, <https://doi.org/10.1016/j.ajog.2020.02.032>.
- [25] W.A. Rocca, C.M. Lohse, C.Y. Smith, J.A. Fields, M.M. Machulda, M.M. Mielke, Association of premenopausal bilateral oophorectomy with cognitive performance and risk of mild cognitive impairment, *JAMA Netw. Open* 4 (2021) 1–13, <https://doi.org/10.1001/jamanetworkopen.2021.31448>.
- [26] J. Ryan, J. Scali, I. Carrière, H. Amieva, O. Rouaud, C. Berr, K. Ritchie, M.-L. Ancelin, Impact of a premature menopause on cognitive function in later life, *BJOG*. 121 (2014) 1729–1739, <https://doi.org/10.1111/1471-0528.12828>.
- [27] W.A. Rocca, C.M. Lohse, C.Y. Smith, J.A. Fields, M.M. Machulda, M.M. Mielke, Association of premenopausal bilateral oophorectomy with cognitive performance and risk of mild cognitive impairment, *JAMA Netw. Open* 4 (2021), e2131448, <https://doi.org/10.1001/jamanetworkopen.2021.31448>.
- [28] C. Fu, W. Hao, N. Shrestha, S.S. Virani, S.R. Mishra, D. Zhu, Association of reproductive factors with dementia: a systematic review and dose-response meta-analyses of observational studies, *EClinicalMedicine*. 43 (2022), 101236, <https://doi.org/10.1016/j.eclinm.2021.101236>.
- [29] V.N. Luine, Estradiol and cognitive function: past, present and future, *Horm. Behav.* 66 (2014) 602–618, <https://doi.org/10.1016/j.yhbeh.2014.08.011>.
- [30] M.T. Weber, P.M. Maki, M.P. McDermott, Cognition and mood in perimenopause: a systematic review and meta-analysis, *J. Steroid Biochem. Mol. Biol.* 142 (2014) 90–98, <https://doi.org/10.1016/j.jsbmb.2013.06.001>.
- [31] S.A. Shumaker, C. Legault, S.R. Rapp, L. Thal, R.B. Wallace, J.K. Ockene, S. L. Hendrix, B.N. Jones 3rd, A.R. Assaf, R.D. Jackson, J.M. Kotchen, S. Wassertheil-Smoller, J. Wactawski-Wende, Estrogen plus progestin and the incidence of dementia and mild cognitive impairment in postmenopausal women: the Women's Health Initiative memory study: a randomized controlled trial, *JAMA*. 289 (2003) 2651–2662, <https://doi.org/10.1001/jama.289.20.2651>.
- [32] H.-H. Zhou, Z. Yu, L. Luo, F. Xie, Y. Wang, Z. Wan, The effect of hormone replacement therapy on cognitive function in healthy postmenopausal women: a meta-analysis of 23 randomized controlled trials, *Psychogeriatrics*. 21 (2021) 926–938, <https://doi.org/10.1111/psyg.12768>.
- [33] Y.-J. Song, S.-R. Li, X.-W. Li, X. Chen, Z.-X. Wei, Q.-S. Liu, Y. Cheng, The effect of estrogen replacement therapy on Alzheimer's disease and Parkinson's disease in postmenopausal women: a meta-analysis, *Front. Neurosci.* 14 (2020) 157, <https://doi.org/10.3389/fnins.2020.00157>.
- [34] W.A. Rocca, B.R. Grossardt, L.T. Shuster, Oophorectomy, estrogen, and dementia: a 2014 update, *Mol. Cell. Endocrinol.* 389 (2014) 7–12, <https://doi.org/10.1016/j.mce.2014.01.020>.
- [35] E. Hogervorst, S. Bandelow, Sex steroids to maintain cognitive function in women after the menopause: a meta-analysis of treatment trials, *Maturitas*. 66 (2010) 56–71, <https://doi.org/10.1016/j.maturitas.2010.02.005>.